Investigating the Effect of Temperature on the Rate Constant of the Decomposition of Aqueous Hydrogen Peroxide.

January 29, 2026

IB Chemistry
Internal Assessment

Word count: 3114 / 3000

1 Introduction

The decomposition of hydrogen peroxide (H_2O_2) to form oxygen and water is a thermodynamically favorable process, but unless catalyzed, it proceeds at an extremely slow rate at room temperature.

$$2\,H_2O_{2(aq)} \longrightarrow 2\,H_2O_{(l)} + O_{2(g)}$$

In this investigation, potassium iodide (KI) was used as a catalyst so that the reaction (specifically, the formation of oxygen gas) inside the system can occur at a measurable rate within a practical time frame. This investigation aims to explore how temperature impacts the rate constant for the catalyzed decomposition reaction of aqeuous hydrogen peroxide. The rate of the reaction is observed indirectly by measuring the initial rate of increase in gas pressure inside a sealed flask, as oxygen gas ($O_{2(g)}$) builds up.

$$2\,H_2O_{2(aq)} \xrightarrow{KI} 2\,H_2O_{(l)} + O_{2(g)} \tag{1}$$

The collision theory states that a chemical reaction can only occur when particles collide with sufficient energy and the correct orientation. As the temperature of the reaction increase, reacting particles have greater average kinetic energy. This increase in the particle's collision ferquency and the increase in the probability of collisions with energy *equal or greater than* the activation energy (E_A). Thus, increasing the temperature of the reaction is expected to increase the rate of the reaction and the reaction constant. The temperature dependence of a rate constant can be modelled by the Arrhenius equation:

$$k = A \cdot e^{-\frac{E_A}{RT}} \tag{2}$$

Where k is the rate constant, A is the frequency factor, E_A is the activation energy, R is the gas constant ($8.31\ \mathrm{J\ K^{-1}\ mol^{-1}}$)[1], and T is the temperature, in Kelvins. Taking the natural logarithm on both sides of the equation will yield a linear relationship:

$$\ln(K) = -\frac{E_A}{RT} + \ln(A) \tag{3}$$

Where the slope of the linear graph is $-\frac{E_A}{R}$, and the y-intercept is $\ln A$. Because this experiment uses pressure measurements rather than concentration measurements, the ideal gas law can be used to relate pressure with the amount of oxygen produced:

$$PV = nRT \tag{4}$$

In a sealed flask, the gas volume is considered as constant. The equation is re-rranged to show that moles of gas

[1] Value obtained from IB Chemistry Data Booklet, version 1.0 (first assessment 2025).

depends on both pressure and temperature:

$$n = \frac{PV}{RT} \tag{5}$$

Then, for a given ΔP, the change in moles is:

$$\Delta n(O_2) = \frac{V_{\text{gas}}}{RT} \cdot \Delta P \tag{6}$$

Where T is the target temperature (K) for a specific trial, and V_{gas} is the gas volume inside the sealed flask ($V_{\text{gas}} = V_{\text{total volume}} - V_{\text{solution}}$).

* * *

As a result, the research question for this investigation is:

How does the temperature affect the rate constant for the KI-catalyzed decomposition of aqueous hydrogen peroxide, determined from the rate of increase of gas pressure in a sealed flask?

* * *

Given that the rate constant of the decomposition of aqueous H_2O_2 is expected change with temperature, increasing T should increase the fraction of particles reacting with enough energy to overcome the E_A, so that a greater proportion of particles can collide, leading to a reaction. Though KI acts as a catalyst (provides an alternative pathway that has a lower E_A), the elementary steps still needs to overcome an E_A. Thus, the rate constant k should increase with temperature, according to Arrhenius's equation (approximate linear relationship of $\ln(k)$ vs. $1/T$).

2 Experimental Design

This experiment was designed to isolate the temperature of the water bath as the sole independent variable affecting the pressure for the KI-catalyzed decomposition reaction of aqueous H_2O_2. Oxygen gas produced during the reaction increased the pressure inside a sealed 250 mL flask. Thus, the increase in the rate of the initial pressure value was used to determine the rate constant at each temperature. A water bath was used to equilibrate the reaction system to within 1°C of the target temperature. The masses of KI were pre-measured and recorded before data collection, and sealed with plastic wrap and rubber band to prevent exposure to moisture.

Independent variable: Temperature of the reaction system. The H_2O_2 in the sealed flask was equilibrated to the target temperature using a water bath before the start of each trial. The temperature was monitored using a

Vernier Temperature Probe; trials were only started once the water bath temperature reached within 1 °C of the target temperature.

Dependent variable: Gas pressure in the sealed flask (kPa).

Controlled variable: Initial amount of reactants, total volume of gas space in the sealed flask, environmental conditions, sampling settings, and apparatus setup were all kept constant. Variables were kept constant to reduce variations in the rate.

2.1 Materials

Apparatus.

- 1000 mL beaker
- (2x) 600 mL beakers
- 250 mL flask
- 250 mL beaker
- 100 mL beaker
- Glass bowl
- Rubber stoppers
- Vernier equipments (Gas Pressure Sensor, Temperature Probe, LabQuest)
- 25 mL Pipette + suction bulb
- 10 mL graduated cylinder
- Digital balance
- Metal stick from ring stand
- Safety goggles
- Lab coat

Reagents.

- 3% hydrogen peroxide, H_2O_2 (aqueous)
- Potassium iodide, KI (solid)
- Distilled water
- Ice

2.2 Procedure

2.2.1 Part I: Setup

1. Obtain safety goggles and lab coat.
2. Gather the required items from *section 2.1* to work station.
3. Connect the gas pressure sensor to the LabQuest, and attach the plastic tubing to the pressure sensor.
4. Connect the other end to the rubber stopper, and close the valve on the stopper.
5. Secure the 250 mL flask on the ring stand using a ring clamp, so that it remains stable throughout data collection.
6. Add 25 mL of distilled water to the 250 mL flask (to represent H_2O_2 during setup).
7. Place the glass bowl on the hot plate and add water so that it submerges the 25 mL water inside the flask.
8. Draw a horizontal line on the neck of the reaction flask to standardize the insertion depth of the rubber stopper in every trial.
   - This ensures that the headspace gas volume (V_{gas}) remain constant between trials.
   - Before inserting the rubber stopper each time, add a few drops of water to the rubber stopper to improve the sealing.
9. Ensure the temperature probe is not touching the bottom of the glass bowl or the sides of the flask. Secure the temperature probe using a ring clamp (to standardize the placement between trials)
10. Set the LabQuest to record at *5 samples / second*, for a total of *60 seconds* for each trial.

2.2.2 Part II: Preparation of materials

1. Pour a large amount of aqueous H_2O_2 into a clean 600 mL beaker to use as the master solution for all trials. Excess can be poured back laterusing a funnel.
2. Prepare potassium iodide portions for each trial:
   (a) Using a digital scale, weigh 0.30 g of solid KI into individual plastic cups.
   (b) Label each cup with its corresponding temperature level and trial number.
   (c) Seal each cup using a piece of plastic film, and secure it with rubber band to reduce exposure to moisture from air.
3. Fill another clean 600 mL beaker with ice. This will be used to lower the temperature of the water bath.

2.2.3 Part III: Calibration

1. Open the valve on the rubber stopper so that the pressure sensor is open to the atmosphere. Record the atmospheric pressure for 10 seconds and record the average pressure (kPa).

2. Wash the 25 mL pipette with aqueous H_2O_2 to avoid contamination, then dispose into the waste beaker.

Note: An airtight-sealing calibration was not conducted because the experiment relies on finding the change in pressure over time. The pressure readings would be consistently lower than the actual pressure in the case of a minor leak, which does not affect the rate of change in pressure.

2.2.4 Part IV: Experiment

1. Adjust the water bath temperature to the target temperature.

2. Pipette 25 mL of H_2O_2 to a clean 250 mL beaker. Once the temperature of the water bath reaches within 1 °C of the target temperature, pour the H_2O_2 into the reaction flask. Keep the reaction flask in the water bath to maintain consistency.

3. Measure 5 mL of distilled water using a 10 mL graduated cylinder, and pour into the plastic cup containing the pre-weighed KI for the trial. Stir to dissolve the KI. Repeat twice, so that there is a total of 10 mL of KI solution.

4. Start the data collection and collect a 3 second baseline with the flask sealed. Then, remove the rubber stopper, and quickly add the KI solution into the reaction flask, and immediately seal the flask.

5. Swirl the flask gently for *5 seconds*, then keep the flask still for the remainder of the run.

6. After data collection ends, save the run, and record the LabQuest run number with the corresponding trial.

7. Dispose the solution into the waste beaker, and rinse the reaction flask once using tap water, then rinse again with distilled water.

8. Repeat *steps 1-7* for each trial at each temperature level, for a total of *40 trials*.

2.2.5 Part V: Cleanup

1. Disessemble the apparatus.

2. Save the file on the LabQuest, and download the data to a computer.

3. Neutralize the waste solution by adding a small amount of sodium thiosulfate, then dispose the waste solution down the drain with running water. Return all equipments.

3 Data Collection

Level	Trial	Target / °C	Run	Hot plate / °C	Start temp /°C (±0.5 °C)	End temp / °C (±0.5 °C)	Mass of KI / g (±0.01 g)
1	1	15	4	45	14.4	15.1	0.30
	2		5		15.4	16.0	0.29
	3		6		14.7	14.0	0.32
	4		7		14.9	15.3	0.32
	5		8		15.32	14.4	0.31
2	1	20	48	55	19.3	18.7	0.30
	2		9		21.4	20.5	0.28
	3		10		21.0	21.3	0.29
	4		11		19.7	19.4	0.31
	5		12		19.3	19.8	0.31
3	1	25	13	65	25.2	25.8	0.29
	2		14		25.3	25.8	0.29
	3		15		25.7	26.7	0.29
	4		16		24.7	24.6	0.28
	5		17		25.4	25.9	0.30
4	1	30	18	85	29.6	30.6	0.33
	2		19		29.1	28.5	0.30
	3		20		30.9	31.5	0.30
	4		21		30.0	30.4	0.30
	5		22		30.8	31.0	0.29
5	1	35	23	100	35.6	35.4	0.31
	2		24		35.3	35.8	0.31
	3		25		35.1	33.3	0.32
	4		26		34.7	35.3	0.31
	5		27		35.5	36.3	0.29
6	1	40	43	125	39.8	40.9	0.28
	2		44		41.3	41.1	0.28
	3		45		40.5	37.3	0.29
	4		46		40.0	41.0	0.28
	5		47		39.1	38.8	0.29
7	1	45	33	160	45.6	44.9	0.29
	2		34		44.5	43.2	0.30
	3		35		44.8	44.8	0.29
	4		36		44.5	44.7	0.29
	5		37		45.1	44.2	0.29
8	1	50	38	200	49.1	49.5	0.31
	2		39		49.8	49.0	0.28
	3		40		49.0	48.1	0.29
	4		41		49.2	49.7	0.29
	5		42		49.7	49.9	0.29

Table 1: Data recorded while conducting the experiment.

3.1 Sources of Uncertainty

To reduce random errors, 5 trials were conducted at each temperature level and averaged. Other variables, such as initial H_2O_2 concentration and volume, mass of KI, apparatus setup, and sealing method were all kept constant across all runs so that differences in the initial pressure-increase rate can be attributed primary to temperature.

4 Data Processing

The raw data file can be found in *section A.1*, as the original data obtained from the LabQuest is 300 rows by 80 columns.

Level	**Trial**	**Target / K**	**Start temp / °C** (±0.5 **°C**)	**End temp / °C** (±0.5 **°C**)	**Average temp / °C** (±0.5 **°C**)	**Volume of H_2O_2 / mL** (±0.06 **mL**)	**Mass of KI / g** (±0.01 **g**)
1	1	288.15	14.4	15.1	14.75	25	0.30
	2		15.4	16	15.70		0.29
	3		14.7	14	14.35		0.32
	4		14.9	15.3	15.10		0.32
	5		15.32	14.4	14.86		0.31
2	1	293.15	19.3	18.7	19.00	25	0.30
	2		21.4	20.5	20.95		0.28
	3		21	21.3	21.15		0.29
	4		19.7	19.4	19.55		0.31
	5		19.3	19.8	19.55		0.31
3	1	298.15	25.2	25.8	25.50	25	0.29
	2		25.3	25.8	25.55		0.29
	3		25.7	26.7	26.20		0.29
	4		24.7	24.6	24.65		0.28
	5		25.4	25.9	25.65		0.30
4	1	303.15	29.6	30.6	30.10	25	0.33
	2		29.1	28.5	28.80		0.30
	3		30.9	31.5	31.20		0.30
	4		30	30.4	30.20		0.30
	5		30.8	31	30.90		0.29
5	1	308.15	35.6	35.4	35.50	25	0.31
	2		35.3	35.8	35.55		0.31
	3		35.1	33.3	34.20		0.32
	4		34.7	35.3	35.00		0.31
	5		35.5	36.3	35.90		0.29
6	1	313.15	39.8	40.9	40.35	25	0.28
	2		41.3	41.1	41.20		0.28
	3		40.5	37.3	38.90		0.29
	4		40	41	40.50		0.28
	5		39.1	38.8	38.95		0.29
7	1	318.15	45.6	44.9	45.25	25	0.29
	2		44.5	43.2	43.85		0.30
	3		44.8	44.8	44.80		0.29
	4		44.5	44.7	44.60		0.29
	5		45.1	44.2	44.65		0.29
8	1	323.15	49.1	49.5	49.30	25	0.31
	2		49.8	49	49.40		0.28
	3		49	48.1	48.55		0.29
	4		49.2	49.7	49.45		0.29
	5		49.7	49.9	49.80		0.29

Table 2: Raw data.

4.1 Uncertainty Propagation

The uncertainty values for temperature and pressure were obtained from official documentations of the equipments, and the smallest scale on the digital balance was used as the uncertainty of mass: ±0.01 g. A 25 mL volunmetric pipette was used to obtain the H_2O_2 for all trials, but the exact volumes delivered were not confirmed with a graduated cylinder. Thus, the pipette's tolerance was used as the volume uncertainty: ±0.06 mL. Time was recorded automatically by the LabQuest, so its uncertainty was assumed negligible relative to pressure variation throughout the experiment.

Tool	Uncertainty
Vernier Stainless Steel Temperature Probe	±0.1 °C at 0 °C, ±0.5 °C at 100 °C
Vernier Gas Pressure Sensor	±4 kPa
Digital Balance	±0.01 g
Volunmetric Pipette (Class A TD)	±0.06 mL

Table 3: Uncertainty by tool.

However, the methodology in which the data were collected likely increased the uncertainties; for example, temperature varied during the 60 s data collection period, and the probe may not have reflected the solution temperature. Moreover, though marked, the manual insertion depth of the stopper could have introduced small headspace variations. Thus, while linear interpolation with $\Delta T(x) = 0.20 + \frac{0.50-0.20}{100} \cdot x$ could have been used as the temperature probe's uncertainty, a conservative ±0.5 °C was used for all temperature measurements.

4.1.1 Uncertainty in oxygen production rate

For each calculated rate, there contains random uncertainties and uncertainties from the instruments. Random uncertainty was propagated using the spread of the five trials' rates at each temperature level using standard deviation, and converted to uncertainty in the mean rate using the standard error of the mean ($\frac{\sigma}{\sqrt{5}}$):

$$\Delta \overline{r}_{\text{rand}} = \frac{\sigma}{\sqrt{n}} \tag{7}$$

Where σ is the standard deviation of the trials. Instrument uncertainty was propagated through the rate calculation:

$$\Delta \overline{r}_{\text{rel, inst}} = \sqrt{(\frac{\Delta V_{\text{gas}}}{V_{\text{gas}}})^2 + (\frac{\Delta \overline{m}}{\overline{m}})^2 + (\frac{\Delta T}{T})^2} \tag{8}$$

$$\Delta \overline{r}_{\text{abs, inst}} = \overline{r} \cdot \Delta \overline{r}_{\text{rel, inst}} \tag{9}$$

The uncertainty of m was taken from the linear regression of the pressure-time data, and the uncertainty of V_{gas} and T were taken from *table 3*. Combined:

$$\Delta \overline{r} = \sqrt{(\Delta \overline{r}_{\text{rand}})^2 + (\Delta \overline{r}_{\text{abs, inst}})^2} \tag{10}$$

4.1.2 Uncertainty in the Arrhenius plot (THIS IS AI - UPDATE LATER)

For Arrhenius analysis, the mean oxygen production rate at each temperature level was used to create a plot of ln(rate) against 1/T (T in kelvins). The vertical uncertainty in ln(rate) was obtained by propagating the rate uncertainty through the natural logarithm. For small relative uncertainties, the uncertainty in ln(rate) can be approximated by:

Uncertainty in ln(rate) $\approx$ Δ(rate) / rate

(If you prefer an exact form, you can also use ln(rate +- Δrate), but the approximation above is acceptable when Δrate is small compared with rate.)

The horizontal uncertainty in 1/T comes from the temperature uncertainty. Since 1/T changes with T, the uncertainty was calculated as:

Uncertainty in $(1/T) = \Delta T/T^2$

where ΔT is +-0.5 °C (same magnitude in K) and T is in kelvins.

The activation energy was determined from the gradient of the best-fit line of ln(rate) vs 1/T. Because gradient = -Ea/R, the activation energy was calculated as:

Ea = -(gradient) * R

The uncertainty in Ea was obtained from the uncertainty in the gradient returned by the linear regression of the Arrhenius plot, multiplied by R.

4.2 Processed Data

Temp / °C	Trial	Average temp / K (±0.5 °C)	Mass of KI / g (±0.01 g)	Start window / s	End window / s	O_2 production rate / mol s^{-1}	Avg. rate / mol s^{-1}	Rate SD / mol s^{-1}
15	1	287.90	0.30	37.8	60.0	$1.40 \cdot 10^{-6}$	$(1.89 \pm 0.18) \cdot 10^{-6}$	$3.69 \cdot 10^{-7}$
	2	288.85	0.29	47.4	60.0	$1.94 \cdot 10^{-6}$		
	3	287.50	0.32	30.0	60.0	$2.41 \cdot 10^{-6}$		
	4	288.25	0.32	32.0	60.0	$1.95 \cdot 10^{-6}$		
	5	288.01	0.31	29.2	60.0	$1.74 \cdot 10^{-6}$		
20	1	292.15	0.30	31.8	60.0	$1.67 \cdot 10^{-7}$	$(1.89 \pm 0.44) \cdot 10^{-6}$	$9.88 \cdot 10^{-7}$
	2	294.10	0.28	28.0	60.0	$2.60 \cdot 10^{-6}$		
	3	294.30	0.29	25.6	60.0	$2.26 \cdot 10^{-6}$		
	4	292.70	0.31	30.8	60.0	$2.44 \cdot 10^{-6}$		
	5	292.70	0.31	32.8	60.0	$1.97 \cdot 10^{-6}$		
25	1	298.65	0.29	29.8	60.0	$3.88 \cdot 10^{-6}$	$(3.39 \pm 0.29) \cdot 10^{-6}$	$6.37 \cdot 10^{-7}$
	2	298.70	0.29	26.6	60.0	$4.19 \cdot 10^{-6}$		
	3	299.35	0.29	34.2	60.0	$3.17 \cdot 10^{-6}$		
	4	297.80	0.28	24.0	60.0	$2.60 \cdot 10^{-6}$		
	5	298.80	0.30	33.8	60.0	$3.12 \cdot 10^{-6}$		
30	1	303.25	0.33	10.0	60.0	$1.10 \cdot 10^{-5}$	$(7.54 \pm 1.20) \cdot 10^{-6}$	$2.67 \cdot 10^{-6}$
	2	301.95	0.30	13.4	60.0	$5.31 \cdot 10^{-6}$		
	3	304.35	0.30	7.4	60.0	$9.73 \cdot 10^{-6}$		
	4	303.35	0.30	17.0	60.0	$6.42 \cdot 10^{-6}$		
	5	304.05	0.29	19.4	60.0	$5.24 \cdot 10^{-6}$		
35	1	308.65	0.31	7.8	60.0	$1.03 \cdot 10^{-5}$	$(8.59 \pm 0.93) \cdot 10^{-6}$	$2.07 \cdot 10^{-6}$
	2	308.70	0.31	16.8	60.0	$7.85 \cdot 10^{-6}$		
	3	307.35	0.32	15.4	60.0	$8.87 \cdot 10^{-6}$		
	4	308.15	0.31	11.6	60.0	$1.05 \cdot 10^{-5}$		
	5	309.05	0.29	20.6	60.0	$5.44 \cdot 10^{-6}$		
40	1	313.50	0.28	11.2	60.0	$9.82 \cdot 10^{-6}$	$(8.89 \pm 1.21) \cdot 10^{-6}$	$2.70 \cdot 10^{-6}$
	2	314.35	0.28	18.6	60.0	$8.19 \cdot 10^{-6}$		
	3	312.05	0.29	16.2	60.0	$6.24 \cdot 10^{-6}$		
	4	313.65	0.28	8.6	60.0	$1.31 \cdot 10^{-5}$		
	5	312.10	0.29	14.6	60.0	$7.10 \cdot 10^{-6}$		
45	1	318.40	0.29	7.0	60.0	$1.14 \cdot 10^{-5}$	$(1.29 \pm 0.27) \cdot 10^{-5}$	$6.06 \cdot 10^{-6}$
	2	317.00	0.30	16.4	60.0	$9.41 \cdot 10^{-6}$		
	3	317.95	0.29	4.6	60.0	$1.78 \cdot 10^{-5}$		
	4	317.75	0.29	8.6	60.0	$2.03 \cdot 10^{-5}$		
	5	317.80	0.29	12.4	60.0	$5.63 \cdot 10^{-6}$		
50	1	322.45	0.31	10.4	60.0	$3.09 \cdot 10^{-5}$	$(1.84 \pm 0.38) \cdot 10^{-5}$	$8.54 \cdot 10^{-6}$
	2	322.55	0.28	11.8	60.0	$1.16 \cdot 10^{-5}$		
	3	321.70	0.29	14.0	60.0	$1.19 \cdot 10^{-5}$		
	4	322.60	0.29	5.4	52.8	$2.38 \cdot 10^{-5}$		
	5	322.95	0.29	15.2	60.0	$1.40 \cdot 10^{-5}$		

Table 4: Processed data.

Note: Trial 5 of 15°C (underlined) was removed from calculations because the source data could not be located on the LabQuest. Run 8 from the LabQuest indicated a target temperature of 293.15K, which contradicts with trial 5's target temperature of 288.15 K.

4.3 Example Calculation

Consider 15 °C trial 1 as an example (marked as red) in *table 4*. The average temperature for this trial will be used to determine ΔT:

$$T = 14.75°\mathrm{C} + 273.15 = 287.9\mathrm{K} \tag{11}$$

The headspace gas volume was calculated as the difference between the flask volume and the solution volume:

$$V_{\text{gas}} = 250 \text{ mL} - 25 \text{ mL} = 225 \text{ mL} \rightarrow 2.25 \cdot 10^{-4} \text{ m}^3 \tag{12}$$

The analysis window for the pressure-time graph was selected as the main region where pressure increased (from the start of sustained pressure increase to the end of the rise). For this trial, the selected interval was 37.8 s to 60 s:

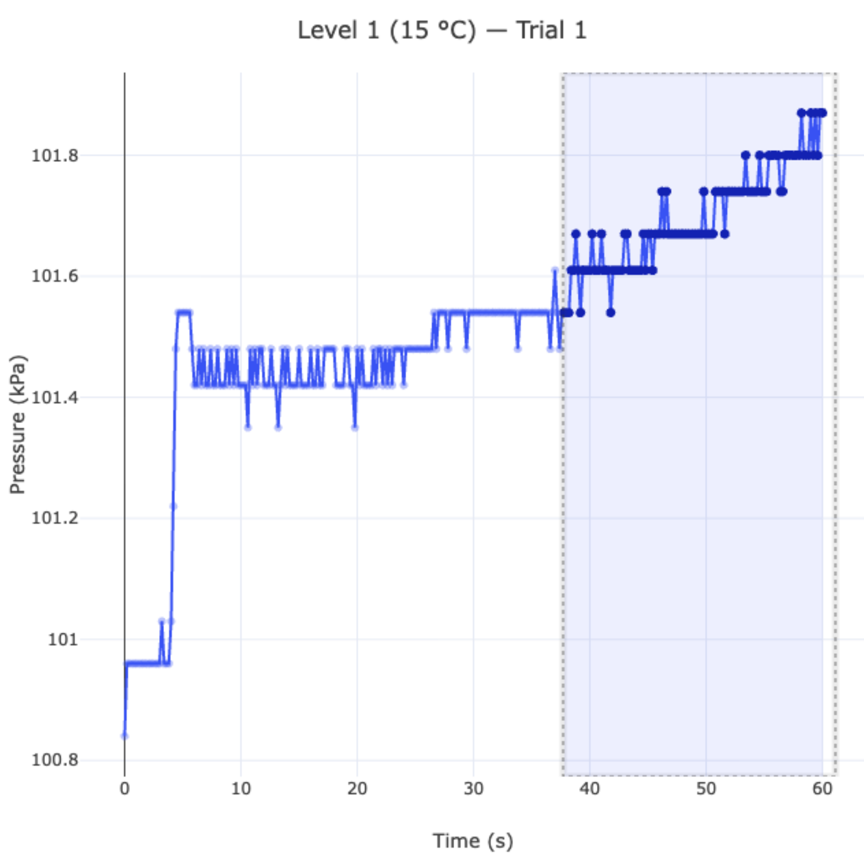

Figure 1: Pressure-time graph.

This window was chosen because the pressure before $t_{\text{start}} = 37.8$ s showed negligible change in pressure, while the pressure after t_{start} increased consistently, due to $O_{2(g)}$ production. A linear regression with $y = mx + b$ was then fitted to the 111 datapoints in the selected window:

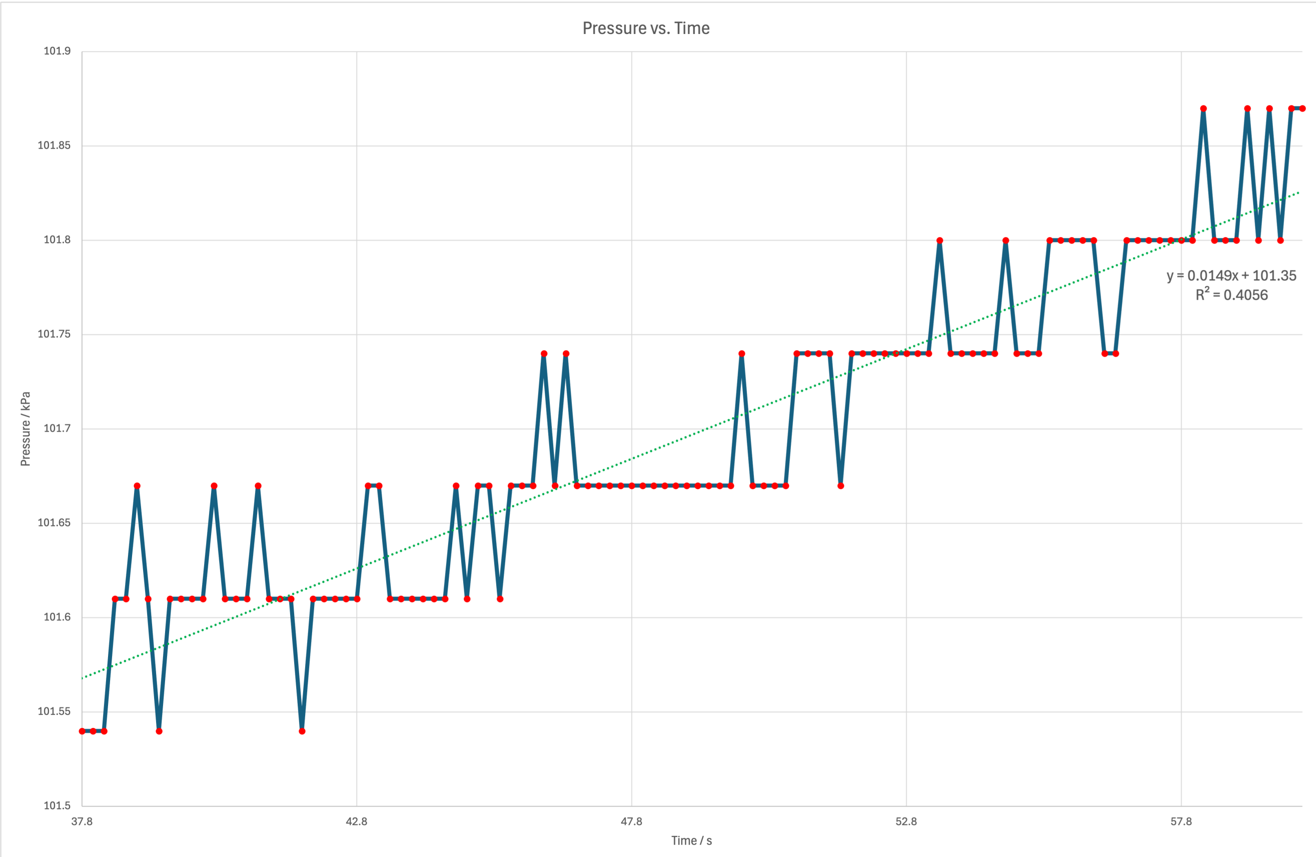

Figure 2: Selected analysis window.

Note that the slope m was recorded as 0.01486 kPa s^{-1}, which converts to 14.86 Pa s^{-1}. The rate of O_2 production was then calculated from the ideal gas relationship. For a constant headspace volume:

$$r = \frac{\Delta n(\mathrm{O_2})}{\Delta t} = \frac{V_{\text{gas}}}{RT} \cdot (\frac{\Delta P}{\Delta t}) = \frac{V_{\text{gas}}}{RT} \cdot m \tag{13}$$

$$r = \frac{2.25 \cdot 10^{-4}\ \mathrm{m^3}}{(8.31\ \mathrm{J\ K\ mol^{-1}})(288.15\ \mathrm{K})} \cdot 14.86\ \mathrm{Pa\ s^{-1}} \tag{14}$$

$$r = 1.40 \cdot 10^{-6}\ \mathrm{mol\ s^{-1}}$$

The process above was repeated to calculated the average rate for the other 4 trials at 15 °C:

$$\overline{r} = \frac{1}{n} \sum_{i=1}^{n=5} r_i \tag{15}$$

$$\overline{r} = \frac{(1.40 \cdot 10^{-6}) + (1.94 \cdot 10^{-6}) + (2.41 \cdot 10^{-6}) + (1.95 \cdot 10^{-6})}{4} = 1.88 \cdot 10^{-6}\ \mathrm{mol\ s^{-1}} \tag{16}$$

Note: Trial 5 was not included in the calculations and justified in *section 4*.

The random uncertainty can then be calculated:

$$\sigma = \sqrt{\frac{1}{4}(-0.48 \cdot 10^{-6})^2 + (0.06 \cdot 10^{-6})^2 + (0.53 \cdot 10^{-6})^2 + (0.07 \cdot 10^{-6})^2} \quad (17)$$

$$\sigma = 3.69 \cdot 10^{-7} \text{ mol s}^{-1}$$

$$\Delta \overline{r}_{\text{rand}} = \frac{\sigma}{\sqrt{4}} = 1.65 \cdot 10^{-7} \text{ mol s}^{-1} \quad (18)$$

Calculating instrumental uncertainty from known variables:

1. Average temperature range in this level: 0.4 °C $\rightarrow$ half-range = 0.2 °C

$$\Delta T = \max(0.5, 0.2) = \pm 0.5 \text{ K}$$

$$\therefore \frac{\Delta T}{T} = \frac{0.5 \text{ K}}{288.15 \text{ K}} = 0.001735 \rightarrow 0.174\% \quad (19)$$

2. Average slope uncertainty (from Microsoft Excel's Regression tool): ± 0.457 Pa s^{-1}

$$\frac{\Delta m}{m} = \frac{0.457 \text{ Pa}}{14.86 \text{ Pa}} = 0.0307 \rightarrow 3.07\% \quad (20)$$

3. Volume uncertainty: ± 0.06 mL:

$$\frac{\Delta V_{\text{gas}}}{V_{\text{gas}}} = \frac{0.06 \text{ mL}}{225 \text{ mL}} = 0.000267 \rightarrow 0.0267\% \quad (21)$$

The instrumental uncertainties are then combined:

$$\Delta \overline{r}_{\text{rel, inst}} = \sqrt{(0.001735)^2 + (0.0307)^2 + (0.000267)^2} = 0.0308 \rightarrow 3.08\% \quad (22)$$

$$\therefore \Delta \overline{r}_{\text{abs, inst}} = \overline{r} \cdot \Delta \overline{r}_{\text{rel, inst}} = (1.88 \cdot 10^{-6}) \cdot 0.0308 = 5.79 \cdot 10^{-8} \text{ mol s}^{-1} \quad (23)$$

Final uncertainty for the average rate since they are independent uncertainties:

$$\Delta \overline{r} = \sqrt{(1.65 \cdot 10^{-7})^2 + (5.79 \cdot 10^{-8})^2} \quad (24)$$

$$\Delta \overline{r} = 1.75 \cdot 10^{-7} \text{ mol s}^{-1}$$

* * *

The approach from above was used to calculate the apparent rate constant for the other 8 temperature levels:

Temperature level / °C	**Temperature level / K**	**Mean O_2 production rate, $\bar{r}$ / mol s^{-1}**
15	288.15	$(1.89 \pm 0.18) \cdot 10^{-6}$
20	293.15	$(1.89 \pm 0.44) \cdot 10^{-6}$
25	298.15	$(3.39 \pm 0.29) \cdot 10^{-6}$
30	303.15	$(7.54 \pm 1.20) \cdot 10^{-6}$
35	308.15	$(8.59 \pm 0.93) \cdot 10^{-6}$
40	313.15	$(8.89 \pm 1.21) \cdot 10^{-6}$
45	318.15	$(1.29 \pm 0.27) \cdot 10^{-5}$
50	323.15	$(1.84 \pm 0.38) \cdot 10^{-5}$

Table 5: Mean oxygen production rate.

5 Data Analysis

Data from *table 4* and *table 5* were used to create an average initial rate (kPa s^{-1}) vs. temperature (K) graph:

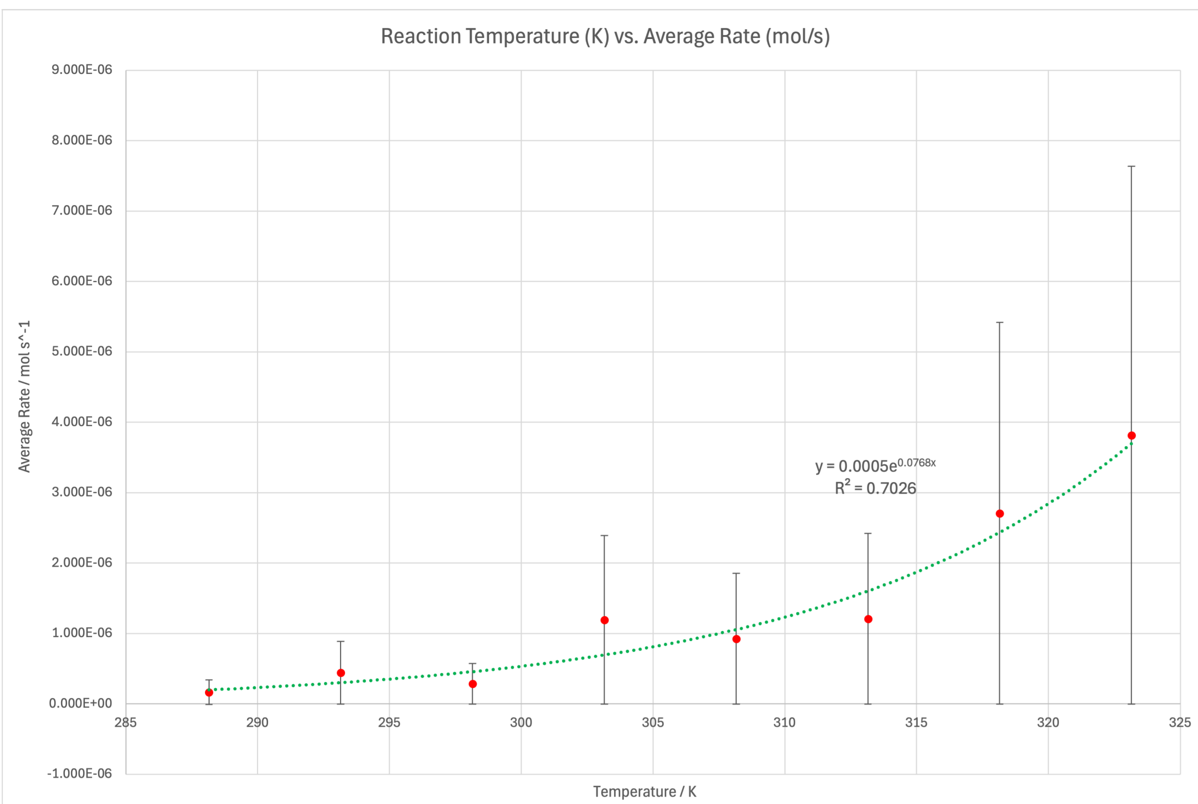

Figure 3: Graph of average initial rate (kPa s^{-1}) vs. temperature level (K).

As shown in *figure 3* above, there is an increasing exponential relationship between the apparent rate constant k_{app} and the temperature of the reaction:

$$f(x) = 0.0005e^{0.0768x} \tag{25}$$

The apparent rate constant k_{app} was defined in *section* **??** to represent the initial rate of the reaction within the first 15 seconds after the reaction begins. Given that the initial amounts of H_2O_2 and I^- were held constant across all trials (with minor variations in the mass of KI), k_{app} is proportional to the actual rate constant k for the reaction, by a constant scale factor difference:

$$k_{app} = C \cdot k, \tag{26}$$

Where C is the constant scaling factor between the apparent rate constant and the true rate constant. As shown in *table 4* and *table 5*, the apparent rate constant increases from an average of 0.01023 ± 0.00194 kPa s^{-1} at 15°C (288.15K) to an average of 0.10081 ± 0.02089 kPa s^{-1} at 50°C (323.15K), which is an increase by a factor of 9.85. This supports my initial hypothesis that higher temperature increases the frequency of collisions and the fraction of molecules with energy $\geq E_A$. To better model how the reaction temperature impacts the rate constant, the Arrhenius equation from *equation 2* was used to analyze the data, given that $k_{app} \propto k$:

$$k_{app} = A_{app} \cdot e^{-\frac{E_A}{RT}} \tag{27}$$

An Arrhenius plot of $\ln(k_{app})$ vs. $\frac{1}{T}$ was created by taking the natural logarithms on both sides:

$$\ln(k_{app}) = -\frac{E_A}{R} \cdot (\frac{1}{T}) + \ln(A_{app}) \tag{28}$$

Recall from *section* **??** that the slope of the Arrhenius plot is $-\frac{E_A}{R}$ and the y-intercept is $\ln(A)$. The Arrhenius equation can then be rearranged, where m is the slope of the Arrhenius plot, and b is the y-intercept:

$$\ln(k_{app}) = m \cdot (\frac{1}{T}) + b \tag{29}$$

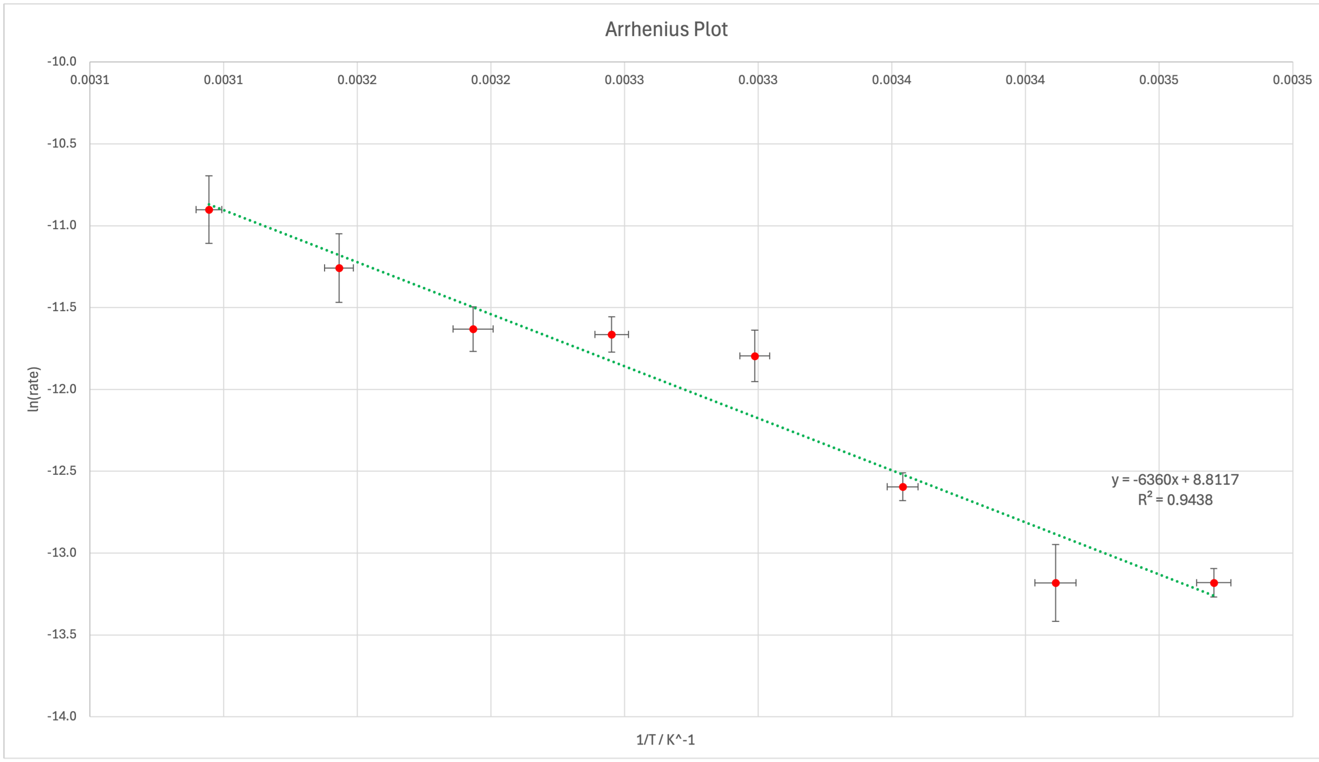

Figure 4: Arrhenius plot, graph of $\ln k_{app}$ vs. $\frac{1}{T} \cdot 10^3$.

However, because the x-axis of the graph uses $\frac{1}{T} \cdot 10^{-3}$ rather than $\frac{1}{T}$, the slope must be multiplied by 1000 to be correct:

$$\ln{(k_{app})} = (m \cdot 1000) \cdot (\frac{1}{T}) + b \tag{30}$$

Plugging in the values from *figure 4*:

$$\ln{(k_{app})} = (-7.2974 \cdot 10^3) \cdot (\frac{1}{T}) + 18.815 \tag{31}$$

Given that the slope $m = -\frac{E_A}{R}$, the activation energy E_A can be calculated using $R = 8.31$ J K^{-1} mol^{-1}[2]:

$$E_A = |-m \cdot R| = 7.2974 \cdot 10^3 \cdot 8.31 \text{ J K}^{-1} \text{ mol}^{-1} = 60641.4 \text{ J mol}^{-1} \tag{32}$$

$$\therefore E_A = 60.6 \pm 9.4 \text{ kJ mol}^{-1} \tag{33}$$

Thus, the E_A for the KI-catalyzed decomposition reaction is 60.6 ± 9.4 kJ mol^{-1}. Note that the uncertainty for activation energy was obtained from Excel's LINEST function.

A graph of the mass of KI vs. k_{app} was also plotted to check whether small variations in the mass of KI had

[2] Value obtained from IB Chemistry Data Booklet, version 1.0 (first assessment 2025).

an effect on the apparent rate constant. However, no clear relationship was observed, which shows that the small variations in the mass of KI was not the primary cause of the observed temperature dependence of k_{app}:

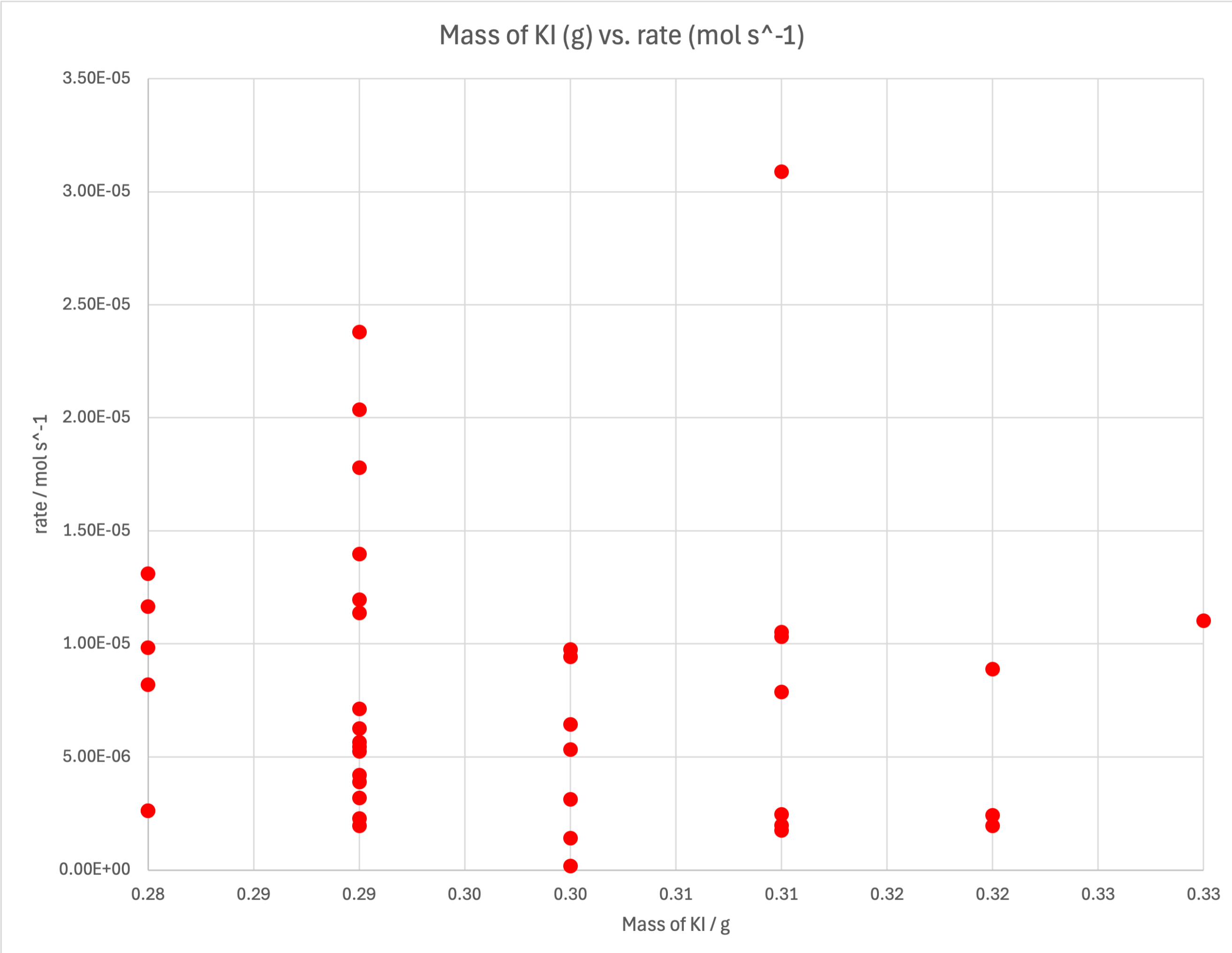

Figure 5: Graph of mass of KI vs. k_{app} for all trials.

6 Conclusion

This investigation aims to examine the relationship between reaction temperature and the initial rate of the KI-catalyzed decomposition reaction of hydrogen peroxide, H_2O_2, where the initial rate (k_{app}) was determined from the initial slope of pressure (kPa) vs. time (s) in a sealed flask. The experimental data showed that higher temperatures increased both the collision frequency and the fraction of molecules with kinetic energy KE $\geq$ E_A, leading to a larger initial pressure-rise rate as oxygen is produced. The hypothesis in *section* **??** predicting an **exponential relationship** between temperature and the apparent rate constant was **supported** by the data.

Moreover, given that the apparent rate constant is proportional to the actual rate constant as apparatus setup and initial reactant amounts were held constant, the Arrhenius plot and graph was used, which resulted in an activation energy of $E_A = 60.6 \pm 9.4$ kJ mol^{-1} for the KI-catalyzed pathway. This value is consistent with accepted scientific values online: a lab published by MIT OpenCourseware and the Chemistry LibreTexts reports an uncatalyzed E_A at about 75 kJ mol^{-1}, while iodide catalysts lowers it to about 56 to 56.5 kJ mol^{-1} [1][2], which is consistent with the experimental data within uncertainty.

Overall, increasing the reaction temperature does increase the measured rate constant for this reaction, as the KI provides an alternative reaction pathway that requires less energy.

7 Evaluation

7.1 Strengths

A key strength was controlling variables.The same apparatus (flask, stopper, tubing, pressure sensor) was used throughout, and the initial reactant amounts were held constant, so changes in the reaction rate are primarily due to temperature. Minor fluctuations in the mass of KI were present, but as *figure 5* has proved, they are not the dominant influence on k_{app}.

Another strength was replication. Learning from past mistakes in other IB Science investigations, conducting five trials per temperature level allowed mean values to be used, reducing random error and making the trend in k_{app} vs. temperature less dependent on a single trial. It also improved stability when a trial and using mean values reduced the impact of random variations between trials and increased stability when a trial needs to be excluded (i.e. due to missing source data or other experimental errors).

7.2 Weaknesses & Limitations

A major methodological limitation was maitaining a stable and uniform water bath temperature during the 60 seconds collection period. While most trials started and ended within 1°C of the target temperature, gradual drift could occur amidst the collection period due to heat loss to surroundings. Moreover, from personal experience, the temperature probe may not have perfectly represented the temperature of the entire water bath; readings often drifted when blocks of ice drifted nearby. To reduce this limitation, a laboratory thermostatic water bath with internal circulation would provide a more uniform temperature control. However, the high cost of such equipment made it impractical for this investigation.

Figure 6: Fisherbrand Isotemp General Purpose Deluxe Water Baths

Figure 7: Thermo Scientific Immersion Circulators

A second limitation was the reaction temperature inside the flask. The solution inside the flask may not have fully equilibrated with the bath during the fitted interval due to the addition of the catalyst (prepared by dissolving solid KI in water at room-temperature), which would increase the spread of data in k_{app} and reduce accuracy of teh temperature assigned to each trial. This can be improved by pre-equilibrating the KI solution in the same bath, and measure the temperature inside the flask using a 3-hole stopper with a sealed port for the temperature probe.

Lastly, adding the catalyst required breifly opening the system. At higher temperatures, the reaction could begin before a complete seal was achieved, which lead to an underestimation of the initial rate. Injecting the catalyst using a syringe would allow for a more consistent start time across trials and a more proper seal on the rubber stopper.

By implementing these improvements in future iterations, a more complete characterization of how the reaction temperature influences the inital reaction rate of the decomposition reaction of hydrogen peroxide.

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A Appendix

A.1 Complete Raw Data

The Microsoft Excel file for the complete raw data can be found at GitHub: `https://github.com/AccessRetrieved/chemistryia/blob/d6167ade96b497181fbf9c2e57b69c0467dfe979/Data/Data.xlsx`.

A.2 Source Code

All source code and raw data are open-source, available to view on GitHub: `https://github.com/AccessRetrieved/chemistryia`.